

1010683 → Rout Dutta

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Date 5/12/2015

ML Assignment-3

Machine Learning
Professional

(1.)

1	1	2	0	3
2	4	2	1	1
1	1	5	3	0
1	2	6	0	1
0	1	2	0	3

(A)

1	0	-1
0	1	0
-1	0	1

= ~~5~~

Stride = 1
Padding = 0

$$S_{11} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 2 \\ 1 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1+0-2 \\ 0+4+0 \\ -1+0+5 \end{bmatrix} \rightarrow 7$$

$$S_{12} = \begin{bmatrix} 1 & 2 & 0 \\ 4 & 2 & 1 \\ 1 & 5 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1+0+0 \\ 0+2+0 \\ -1+5+3 \end{bmatrix} \rightarrow 5$$

$$S_{13} = \begin{bmatrix} 2 & 0 & 3 \\ 2 & 1 & 1 \\ 5 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2+0-3 \\ 0+1+0 \\ -5+0+0 \end{bmatrix} \rightarrow -5$$

$$S_{21} = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 1 & 5 \\ 1 & 2 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2+0-2 \\ 0+1+0 \\ -1+0+6 \end{bmatrix} \rightarrow 6$$

$$S_{22} = \begin{bmatrix} 4 & 2 & 1 \\ 1 & 5 & 3 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 4+0-1 \\ 0+5+0 \\ -2+0+0 \end{bmatrix} \rightarrow 6$$

$$S_{23} = \begin{bmatrix} 2 & 1 & 1 \\ 5 & 3 & 0 \\ 6 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2+0-1 \\ 0+3+0 \\ -6+0+1 \end{bmatrix} \rightarrow -1$$

$$S_{31} = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 2 & 6 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1+0+(-5) \\ 0+2+0 \\ 0+0+2 \end{bmatrix} \rightarrow 0$$

$$S_{32} = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 6 & 0 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1+0+(-3) \\ 0+6+0 \\ -1+0+0 \end{bmatrix} \rightarrow 3$$

$$S_{33} = \begin{bmatrix} 5 & 3 & 0 \\ 6 & 0 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 5+0+0 \\ 0+0+0 \\ -2+0+3 \end{bmatrix} \rightarrow 6$$

$$S = \begin{bmatrix} 7 & 5 & 5 \\ 6 & -1 \\ 0 & 3 & 6 \end{bmatrix}$$

ANSWER

(K) Padding = 1, stride = 2, ReLU = $\max(0, x)$

0	0	0	0	0	0	0
0	1	1	2	0	3	0
0	2	4	2	1	1	0
0	1	1	5	3	0	0
0	1	2	6	0	1	0
0	0	1	2	0	3	0
0	0	0	0	0	0	0

1	0	-1
0	1	0
-1	0	1

Here, $|x| \rightarrow \text{ReLU}(x)$

$$\text{ReLU}(S_1) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = |5| = 5$$

$$\text{ReLU}(S_2) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 2 & 0 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} = |-1| = 1$$

$$\text{ReLU}(S_3) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} = |2| = 2$$

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$$\text{ReLU}(\xi_{21}) = \begin{bmatrix} 0 & 2 & 4 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 10 & -1 \\ 0 & 1 & 0 \\ -10 & 1 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$\text{ReLU}(\xi_{31}) = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 10 & -1 \\ 0 & 1 & 0 \\ -10 & 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

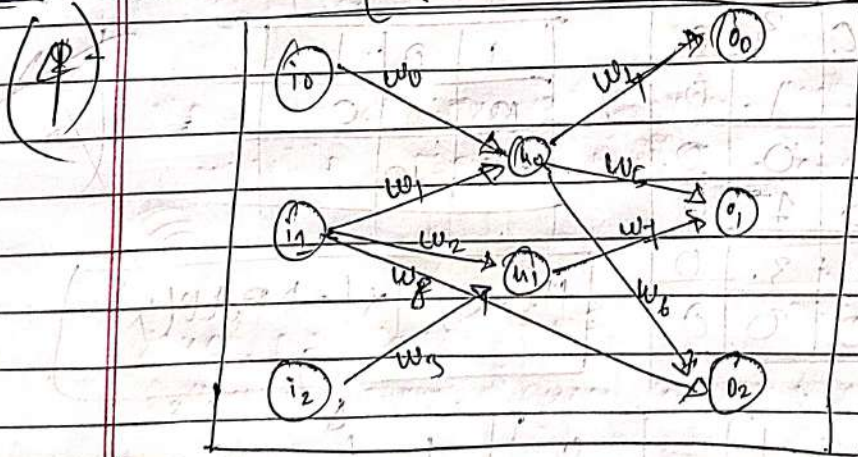
$$\text{ReLU}(\xi_{22}) = \begin{bmatrix} 4 & 2 & 1 \\ 1 & 5 & 3 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 10 & -1 \\ 0 & 1 & 0 \\ -10 & 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ -2 \end{bmatrix} \Rightarrow \begin{bmatrix} 9 \\ 5 \\ 0 \end{bmatrix}$$

$$\text{ReLU}(\xi_{32}) = \begin{bmatrix} 2 & 6 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 10 & -1 \\ 0 & 1 & 0 \\ -10 & 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

$$\text{ReLU}(\xi_{23}) = \begin{bmatrix} 1 & 0 \\ 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 10 & -1 \\ 0 & 1 & 0 \\ -10 & 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 10 \\ 3 \\ 0 \end{bmatrix}$$

$$\text{ReLU}(\xi_{33}) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 10 & -1 \\ 0 & 1 & 0 \\ -10 & 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$$

$$\text{ReLU}(\eta) = \begin{bmatrix} 5 & 0 & 2 \\ 0 & 6 & 1 \\ 0 & 4 & 3 \end{bmatrix}$$



$$a_{h0} = w_0 x_0 + w_1 x_1 : h_0 = \sigma(w_0 x_0 + w_1 x_1)$$

$$a_{h1} = w_2 x_1 + w_3 x_2 : h_1 = \sigma(w_2 x_1 + w_3 x_2)$$

Incoming edges to o_2 : from i_1 via w_8 , from h_0 via w_7

$$\therefore a_{o2} = w_8 x_1 + w_7 h_0 : o_2 = \sigma(w_8 x_1 + w_7 \sigma(w_0 x_0 + w_1 x_1))$$

(b) In Neural Nets, weights are learnable params. They are learned via GPO.

For that we must calculate, weight gradients wrt the loss function.

$w_{ij} \Rightarrow$ (weight of graph edge going $i \rightarrow j$).

$$\frac{\partial L}{\partial w_{ij}} = \frac{\partial L}{\partial a_j} \frac{\partial a_j}{\partial w_{ij}} = \delta_j \cdot x_i$$

δ_j $\rightarrow$ output of previous node
 δ_j $\rightarrow$ error signal of node 'j'

$\delta_j = \frac{\partial L}{\partial a_j} = \sigma'(a_j) \sum_k w_{jk} \delta_k$ Back prop moves in opposite direction.

w_j is an outgoing edge NOT incoming edge. At best it will be from no back prop.

$$\delta_0 = \sigma'(a_0) \cdot [w_4 \delta_0 + w_5 \delta_1 + w_6 \delta_2] \quad \left(\because \frac{\partial L}{\partial a_j} = \sigma'(a_j) \sum_k w_{jk} \delta_k \right)$$

(b) $\frac{\partial L}{\partial w_{ij}}$ [backpropagation to learn weights].

$$= \frac{\partial L}{\partial a_j} \frac{\partial a_j}{\partial w_{ij}} = \delta_j \cdot x_i$$

δ_j $\rightarrow$ output of previous node.
 δ_j $\rightarrow$ error signal of node 'j'.

Now, therefore: $\delta_j = \frac{\partial L}{\partial a_j} = \sigma'(a_j) \left(\sum_k w_{jk} \delta_k \right)$

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weight in question: w_1 [connected to i_1 & h_0].

$$\delta_{h_0} = \sigma'(h_0) \cdot (w_4 \delta_0 + w_5 \delta_1 + w_6 \delta_2).$$

Input nodes do NOT have ERROR signals

$\therefore$ NO δ term contains w_1 term. (ANSWER)

$$w_1' \text{ (weight update formula)} = w_1 - \eta \frac{\partial L}{\partial w_1}.$$

$$\therefore w_1' = w_1 - \eta (\delta_{h_0} \cdot x_1).$$

$$\text{Similarly } w_0' = w_0 - \eta (\delta_{h_0} \cdot x_0).$$

(a) Is h_0 d-sep from h_1 ?

No conditioning given, we have ONE

visible open path: $h_0 \leftarrow i_1 \rightarrow h_1$

Since, At least ONE

TRUNK: open.

path is open, hence h_0 & h_1 are NOT d-separated.

(a) Is i_1 d-sep from o_1 ?

No.

Open path: $i_1 \rightarrow h_0 \rightarrow o_1$, $i_1 \rightarrow h_1 \rightarrow o_1$

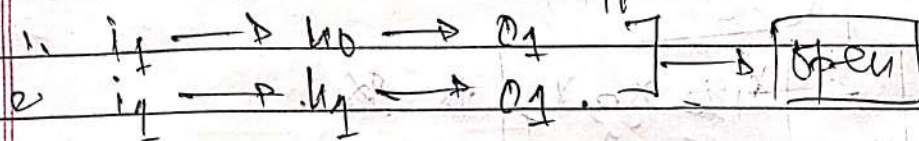
is it [NOT d-separated] from o_1 .

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(e) Nothing changes. weights are NOT probabilistic random variable ~~variables~~. conditioning.

The only valid conditioning is of O_2 .

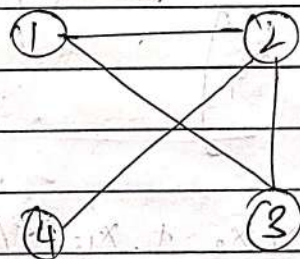
But this does not affect, the already open paths



Hence, i_1 & O_1 are STILL NOT d-separated.

(2)

Fig.



$$Y \rightarrow (y_1, y_2, y_3, y_4)$$

$$0 \leq y_i \leq 4$$

$$X \rightarrow (x_1, x_2, x_3, x_4)$$

$$0 \leq x_i \leq 4$$

$$f_X(x) = \frac{1}{Z_1} \exp(x_1 x_2 x_3 x_4)$$

$$Z_1 = \int_0^4 \int_0^4 \int_0^4 \int_0^4 \exp[x_1 x_2 x_3 x_4]$$

$$f_Y(y) = \frac{1}{Z_2} \exp(y_1 y_2 y_3 + y_1 y_2 + y_4)$$

$$Z_2 = \int_0^4 \int_0^4 \int_0^4 \int_0^4 \exp[y_1 y_2 y_3 + y_1 y_2 + y_4]$$

(a) A random vector $Z = [Z_1, Z_2, \dots, Z_i, \dots, Z_n]$ is a MRF w.r.t G if its density can be written as a product of clique potentials.

$$p(Z) = \prod_{c \in \mathcal{C}(G)} \phi_c(Z_c)$$

c is a clique oriented.
 $\mathcal{C}(G)$ is set of all cliques

$\therefore$ Equivalently in log form: $\log[p(Z)] = \sum_c \phi_c(Z_c) + K$

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cliques of G (given) (i) 1-2-3 [complete closed subgraph]

(ii) 2-4 [2-clique]

Max cliques: $\{1, 2, 3\}, \{2, 4\}$.

In, $f(x) \propto \exp(x_1 x_2 x_3 x_4)$

$$\log[f(x)] = x_1 x_2 x_3 x_4 + \psi$$

(i) 3-4 Not adjacent

(ii) NOT a clique or subclique.

In $f(x) \propto \exp[x_1 x_2 x_3 + x_1 x_2 + x_4]$

$$\log[f(x)] = x_1 x_2 x_3 + x_1 x_2 + x_4 + \psi$$

full maximal clique

$\{x_1, x_2, x_3\}$

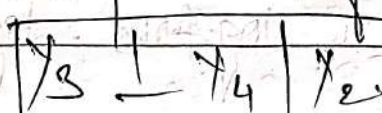
$\{x_2, x_4\}$

VALID!

Terms should be subsets of allowable cliques, in this case $\{x_1, x_2, x_3\}$ & $\{x_2, x_4\}$.

$\therefore \gamma \rightarrow$ valid MRF for G .

(b) By the Markov property, for a random field $(\gamma_1, \gamma_2, \gamma_3, \gamma_4)$ that is Markov with respect to G , separation of 3 & 4 by 2, independence.



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(c) Conditional independence:

In graph G , node 4 has only one neighbour, node 2, since the only edge incident to 4 is $(2,4)$. Therefore, neighbour set of node (4) : $N(4) = 2, \{2,4\}$
Non neighbours of node (4) : $\bar{N}(4) = 2, \{1,3\}$.

The local Markov property states that, for a random field (Y_1, Y_2, Y_3, Y_4) that is Markov w.r.t G , each variable is conditionally independent of all non-neighbours given its neighbours. Hence, for node 4 we obtain the conditional independence.

$$Y_4 \perp (Y_1, Y_3) \mid Y_2$$

(d) $Y_3 = \varepsilon_2$ $\rightarrow$ No parents. (ROOT).
 $Y_1 = 2Y_3 + \varepsilon_1$ $\rightarrow$ Y_3 parent of Y_1 .
 $Y_2 = 4Y_1 + Y_3 + \varepsilon_2$ $\rightarrow$ Y_1, Y_3 parent of Y_2 .
 $Y_4 = 3Y_2 + \varepsilon_4$ $\rightarrow$ Y_2 parent of Y_4 .

$$Y_3 \rightarrow Y_1$$

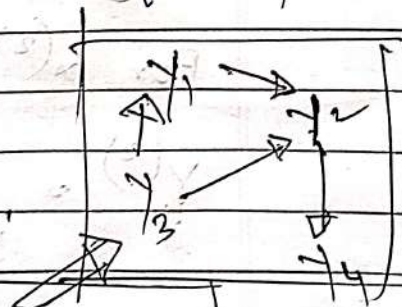
$$Y_1 \rightarrow Y_2, Y_3 \rightarrow Y_2$$

$$Y_2 \rightarrow Y_4$$

$\therefore$ Network:

$$Y_3 \rightarrow Y_1 \rightarrow Y_2$$

$$Y_2 \rightarrow Y_4$$



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- (3.) Two Gaussians $\mathcal{N}(\mu_1, \sigma_1^2), f_1(x), \text{prior} = \pi_1$
 (a) $\mathcal{N}(\mu_2, \sigma_2^2), f_2(x), \text{prior} = \pi_2$.

For 2 component gaussian mixture, the posterior probability of that pt $x^{(i)}$ came from component k is

$$P(z_k^{(i)} = 1 | x^{(i)}) = \gamma_k^{(i)} = \frac{\pi_k f_k(x^{(i)})}{\pi_1 f_1(x^{(i)}) + \pi_2 f_2(x^{(i)})}$$

For $x^{(1)} = 1$.

$$\gamma_1^{(1)} = \frac{\pi_1 f_1(x^{(1)})}{\pi_1 f_1(x^{(1)}) + \pi_2 f_2(x^{(1)})} = \frac{\frac{1}{2} \cdot 0.241}{\frac{1}{2} \cdot 0.241 + \frac{1}{2} \cdot 0.12} \approx \boxed{0.501}$$

$$\gamma_2^{(1)} = 1 - \gamma_1^{(1)} = \boxed{0.499}$$

For $x^{(2)} = 1.5$.

$$\gamma_1^{(2)} = \frac{\frac{1}{2} \cdot 0.13}{\frac{1}{2} \cdot 0.13 + \frac{2}{2} \cdot 0.15} \approx \boxed{0.302}$$

$$\gamma_2^{(2)} = 1 - \gamma_1^{(2)} = \boxed{0.698}$$

$$\begin{aligned} P(z_1^{(1)} = 1 | x^{(1)} = 1) &\approx 0.501 \\ P(z_2^{(1)} = 1 | x^{(1)} = 1) &\approx 0.499 \\ P(z_1^{(2)} = 1 | x^{(2)} = 1.5) &\approx 0.302 \\ P(z_2^{(2)} = 1 | x^{(2)} = 1.5) &\approx 0.698 \end{aligned}$$

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(A) we update mixture weights & means using EM m-step formulas:

$$\pi_k^{new} = \frac{1}{n} \sum_{i=1}^n \gamma_k^{(i)} \quad , \quad \mu_k^{new} = \frac{\sum_{i=1}^n \gamma_k^{(i)} x^{(i)}}{\sum_{i=1}^n \gamma_k^{(i)}}$$

$$\begin{aligned} \gamma_1^{(1)} &= 0.501 & \gamma_2^{(1)} &= 0.499 \\ \gamma_1^{(2)} &= 0.302 & \gamma_2^{(2)} &= 0.698 \end{aligned} \quad \left. \begin{array}{l} \gamma_1^{(i)} \\ \gamma_2^{(i)} \end{array} \right\} \rightarrow \text{Data pt. } i$$

Mixture weights:

$$\pi_1^{(new)} = \frac{1}{2} (0.501 + 0.302) \approx 0.4015$$

$$\pi_2^{(new)} = \frac{1}{2} (0.499 + 0.698) = 0.5985$$

Means:

$$\mu_1^{new} = \frac{1 \cdot 0.501 + 1.5 \cdot 0.302}{0.501 + 0.302} = 1.189$$

$$\mu_2^{new} = \frac{1 \cdot 0.499 + 1.5 \cdot 0.698}{0.499 + 0.698} = 1.291$$

$\pi_1^{new} = 0.4015$	$\pi_2^{new} = 0.5985$
$\mu_1^{new} = 1.189$	$\mu_2^{new} = 1.291$

(a)
$$\sigma_k^{2, new} = \frac{\sum_{i=1}^n \gamma_k^{(i)} (x^{(i)} - \mu_k^{new})^2}{\sum_{i=1}^n \gamma_k^{(i)}}$$

First we calc. $\sigma_1^{2, new}$
Then we calc $\sigma_2^{2, new}$

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For μ_{new}
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$$\text{Numerator} = Y_1^{(1)} (x_1^{(1)} - \mu_{new})^2 + Y_1^{(2)} (x_1^{(2)} - \mu_{new})^2$$

$$= 0.0361 (1 - 1.19)^2 + 0.0961 (1.5 - 1.19)^2$$

$$= 0.0361 (0.501)^2 + 0.0961 (0.302)^2$$

$$= 0.0471$$

$$\text{denom} = S_1 = 0.003$$

$$(\sigma^2)_{new} = \frac{0.0471}{0.803} = 0.0586$$

For $(\sigma^2)_2$:

$$\text{Num} \rightarrow Y_2^{(1)} (x_2^{(1)} - \mu_{new})^2 + Y_2^{(2)} (x_2^{(2)} - \mu_{new})^2$$

$$= 0.0841 (1 - 1.29)^2 + 0.0441 (1.5 - 1.29)^2$$

$$= 0.0841 (0.499)^2 + 0.0441 (0.698)^2$$

$$= 0.0727$$

$$\text{denom} \rightarrow S_2 = 1.137$$

$$\therefore (\sigma^2)_2 = \frac{0.0727}{1.137} = 0.0647$$